

5.3.9.3 Low-speed internal (LSI) RC oscillator

Table 39. LSI oscillator characteristics

Symbol	Parameter	Conditions	Min	Typ	Max	Unit
f_{LSI}	LSI frequency	$V_{DD} = 3.3\text{ V}$, $T_J = 25^\circ\text{C}$	31.4 ⁽¹⁾	32	32.6 ⁽¹⁾	kHz
		$T_J = -40\text{ to }130^\circ\text{C}$	29.4 ⁽²⁾		33.6 ⁽²⁾	
		$T_J = -40\text{ to }140^\circ\text{C}$	28.6 ⁽²⁾	-	33.6 ⁽²⁾	
$t_{su(LSI)}$ ⁽³⁾	LSI oscillator startup time	-	-	80	130	μs
$t_{stab(LSI)}$ ⁽³⁾	LSI oscillator stabilization time (5% of final value)	-	-	120	170	
$I_{DD(LSI)}$ ⁽³⁾	LSI oscillator power consumption	-	-	130	280	nA

1. Guaranteed by test production.
2. Evaluated by characterization - Not tested in production.
3. Specified by design - Not tested in production.

5.3.10 Flash memory characteristics

Table 40. Flash memory characteristics

Specified by design and not tested in production.

Symbol	Parameter	Conditions	Min	Typ	Max	Unit
I_{DD}	Supply current	Word program	-	1	-	mA
		Page erase	-	0.8	-	
		Mass erase	-	0.8	-	

Table 41. Flash memory programming

Symbol	Parameter	Conditions	Min ⁽¹⁾	Typ	Max ⁽¹⁾	Unit
t_{prog}	Word programming time	128 bits (user area)	-	20.0	160.0	μs
		16 bits (OTP / EDATA area)	-	20.0	160.0	
$t_{ERASE\ 8KB}$	Page (8 KB) erase time	-	-	2.0	2.1	ms
$t_{ERASE\ 2KB}$	Page (2 KB) erase time	-	-	2.0	2.1	
t_{ME}	Bank Mass erase time	-	-	96.0		
	Mass erase time	-	-	192.0	200.0	
V_{prog}	Programming voltage	-	2.65	-	3.6	V

1. Evaluated by characterization - Not tested in production.

Table 42. Flash memory user and EDATA endurance and data retention

Symbol	Parameter	Conditions	Min ⁽¹⁾	Unit
N_{END}	Endurance	$T_J = -40\text{ to }140^\circ\text{C}$	10	Kcycle
t_{RET}	Data retention	1 Kcycle at $T_J = 125^\circ\text{C}$	10	Year
		1 Kcycle at $T_J = 85^\circ\text{C}$	30	
		10 Kcycle at $T_J = 55^\circ\text{C}$	30	

1. Evaluated by characterization - Not tested in production.